

# DANDI:000017: Parameters and Data

**Distributed coding of choice, action and engagement across the mouse brain** (Steinmetz, Zatka-Haas, Carandini & Harris, *Nature* 576, 266–273, 2019). Dandiset version 0.240329.1926, converted from NWB to CSV/Parquet with `dandi_export/export_000017.py`.

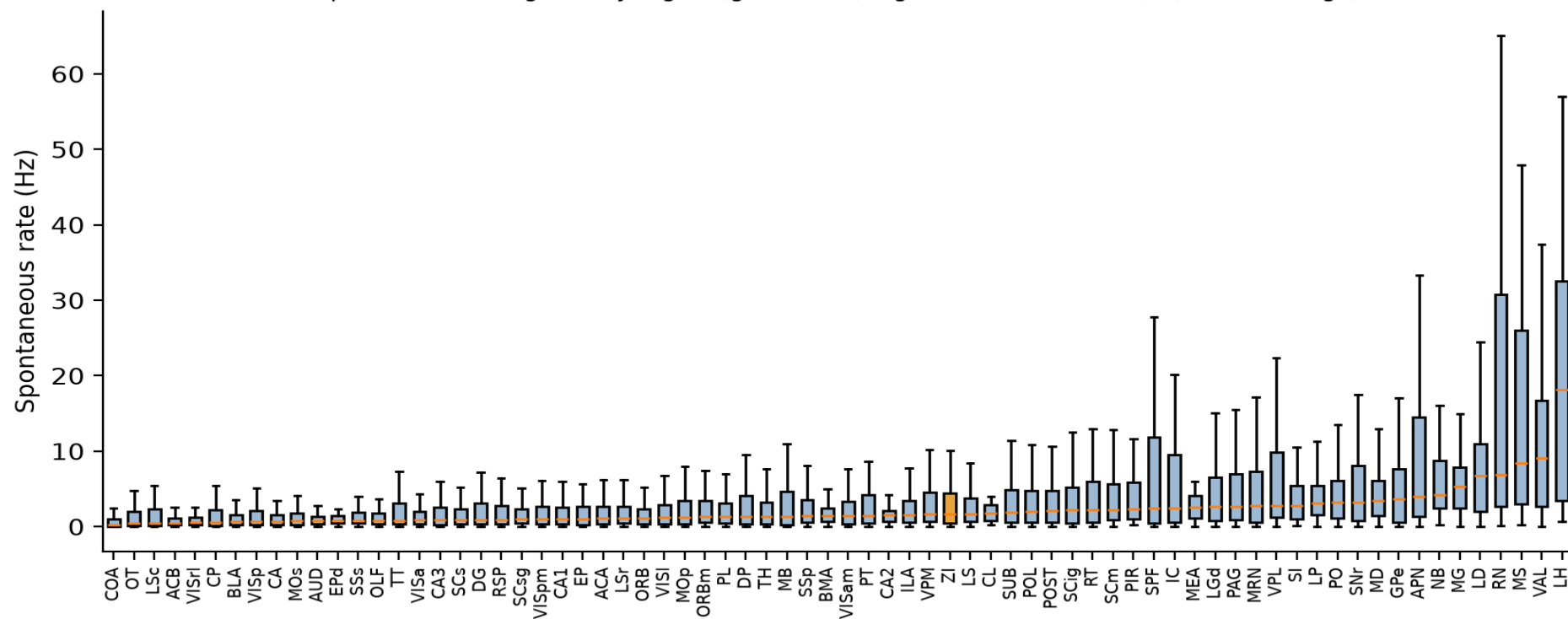
| Quantity                                         | Value           |
|--------------------------------------------------|-----------------|
| Sessions / mice                                  | 39 / 10         |
| Trials (included)                                | 10,050 (9,538)  |
| Sorted units (good, annotation $\geq 2$ )        | 48,661 (33,997) |
| Brain regions (excluding 'root')                 | 72              |
| Total spikes                                     | 442,033,914     |
| Total recording time                             | 35.1 h          |
| ZI good units (sessions)                         | 289 (4)         |
| LHA (lateral hypothalamus) good units (sessions) | 0 (0)           |

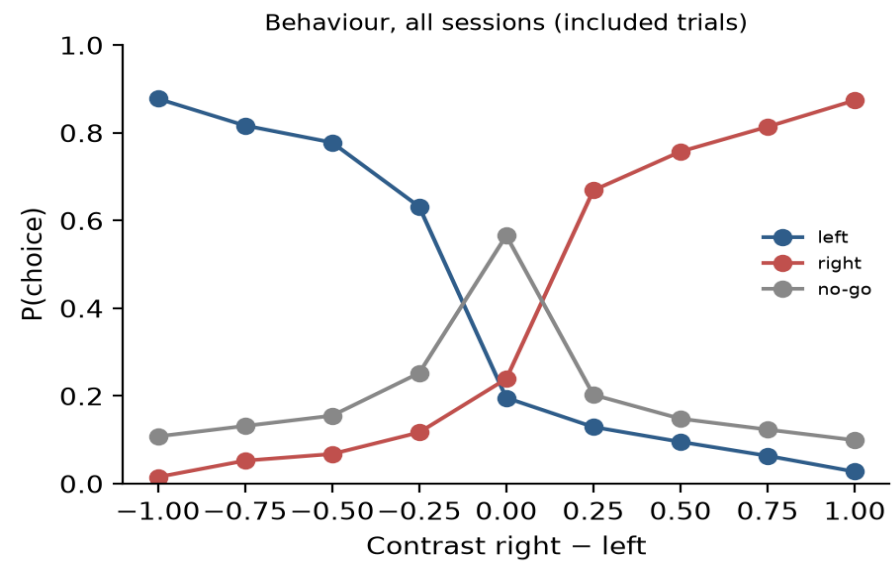
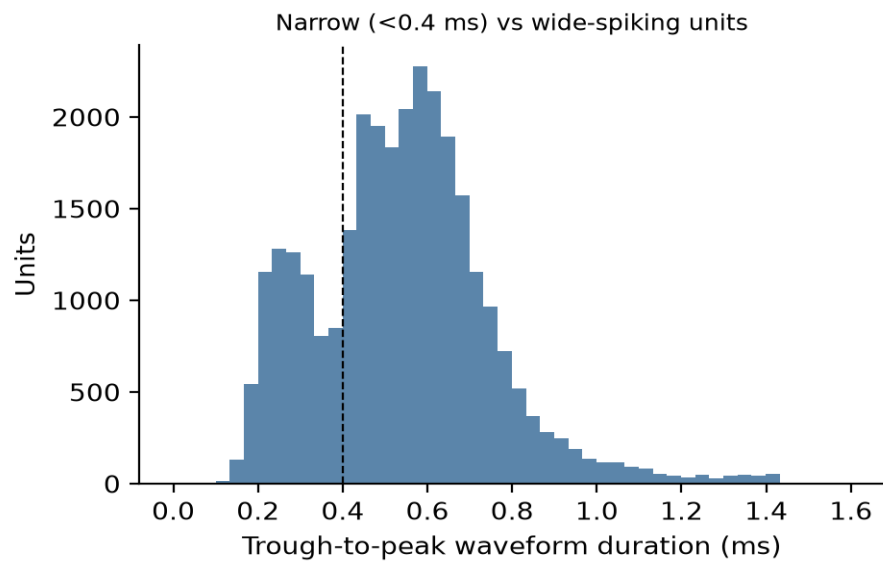
## 1. Parameters

| Group       | Parameter                     | Value                                                                                                                           |
|-------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Subjects    | Species / strain              | Mus musculus; C57Bl6/J background                                                                                               |
| Subjects    | Sex                           | F: 6, M: 4                                                                                                                      |
| Subjects    | Genotypes                     | Ai95 x VGlut1-Cre; C57BL/6J; Snap25-GCaMP6s; Vglut1-Cre; tetO-G6s x CaMK-tTA                                                    |
| Subjects    | Age range                     | 77 – 322 days                                                                                                                   |
| Recording   | Probes                        | Neuropixels Phase 3A, 1–3 probes per session, 384 channels each                                                                 |
| Recording   | Spike sampling rate           | 30 kHz                                                                                                                          |
| Recording   | Spike sorting                 | Kilosort + manual curation in Phy (0 noise excluded; 1 MUA; 2 good; 3 unsorted)                                                 |
| Recording   | Anatomy                       | Probe tracks registered to the Allen Common Coordinate Framework (CCF); region = CCF acronym at the unit's peak channel         |
| Task        | Paradigm                      | 2AFC visual contrast discrimination with no-go option; mouse turns a wheel to bring the higher-contrast grating to centre       |
| Task        | Contrasts (each side)         | 0, 0.25, 0.5, 1                                                                                                                 |
| Task        | Trial events                  | visual stimulus onset → auditory go cue → wheel response → feedback (water reward or white-noise burst)                         |
| Task        | Choice coding                 | +1 left, –1 right, 0 no-go; feedback +1 reward, –1 noise                                                                        |
| Spontaneous | Definition                    | Intervals with no task or stimulus presentation running                                                                         |
| Spontaneous | Duration per session          | median 882 s (range 234–1261 s)                                                                                                 |
| Behaviour   | Signals                       | wheel position (2500 Hz → downsampled to 100 Hz), wheel movements, licks, pupil area/position, face motion energy               |
| Export      | Spike-count window 'prestim'  | -0.5 to +0 s relative to visual_stimulus_time                                                                                   |
| Export      | Spike-count window 'stim'     | +0 to +0.25 s relative to visual_stimulus_time                                                                                  |
| Export      | Spike-count window 'premove'  | -0.25 to +0 s relative to response_time                                                                                         |
| Export      | Spike-count window 'feedback' | +0 to +0.5 s relative to feedback_time                                                                                          |
| Export      | Firing rates                  | mean = spikes / whole recording; task = within first-trial-start ... last-trial-end; spontaneous = within spontaneous intervals |
| Export      | Waveform duration             | trough-to-peak of the mean extracellular waveform (samples / 30 kHz). Not equivalent to intracellular AP half-width             |

## 2. Overview figures

Spontaneous firing rate by region (good units; regions with  $\geq 20$  units; ZI/LHA in orange)





### 3. Firing rates by brain region (good units)

Rows for regions studied in the literature extraction (ZI = zona incerta, LHA = lateral hypothalamic area) are highlighted. Allen 'LH' is the lateral habenula. Spontaneous rate is the value most directly comparable to the literature 'Spontaneous firing rate in Hz' field. Note that these are extracellular recordings in awake behaving mice, not slice patch-clamp.

| Region | Units | Sessions | Spont. mean (Hz) | Spont. SD (Hz) | Spont. median (Hz) | Overall mean (Hz) | Waveform dur. median (ms) |
|--------|-------|----------|------------------|----------------|--------------------|-------------------|---------------------------|
| MOS    | 1743  | 18       | 1.51             | 2.36           | 0.74               | 1.43              | 0.63                      |
| CA1    | 1360  | 21       | 2.19             | 3.39           | 1.03               | 2.18              | 0.63                      |
| TH     | 1257  | 8        | 2.77             | 4.10           | 1.36               | 2.75              | 0.43                      |
| CP     | 1215  | 5        | 2.42             | 4.53           | 0.59               | 2.51              | 0.63                      |
| VISp   | 1118  | 12       | 1.88             | 3.04           | 0.70               | 2.03              | 0.60                      |
| MRN    | 1019  | 11       | 7.06             | 11.65          | 2.73               | 7.41              | 0.30                      |
| VISam  | 1001  | 11       | 2.86             | 4.02           | 1.46               | 2.88              | 0.60                      |
| LGd    | 882   | 10       | 4.79             | 6.59           | 2.62               | 4.44              | 0.47                      |
| MB     | 874   | 6        | 5.45             | 11.45          | 1.38               | 5.07              | 0.33                      |
| PL     | 863   | 10       | 2.64             | 3.73           | 1.31               | 2.42              | 0.63                      |
| ACA    | 841   | 11       | 2.09             | 2.83           | 1.07               | 1.88              | 0.63                      |
| ORB    | 813   | 5        | 1.86             | 2.50           | 1.12               | 1.67              | 0.60                      |
| SCig   | 812   | 4        | 4.38             | 7.43           | 2.19               | 4.53              | 0.27                      |
| MOP    | 793   | 3        | 3.26             | 5.25           | 1.28               | 3.56              | 0.57                      |
| DG     | 776   | 16       | 3.08             | 6.64           | 0.95               | 3.26              | 0.50                      |
| RSP    | 770   | 9        | 2.03             | 2.97           | 0.96               | 1.86              | 0.57                      |
| LP     | 748   | 11       | 4.26             | 4.36           | 3.16               | 4.04              | 0.50                      |
| PO     | 692   | 5        | 4.18             | 4.10           | 3.19               | 4.47              | 0.47                      |
| SUB    | 685   | 9        | 3.90             | 5.41           | 1.85               | 3.91              | 0.57                      |
| VISpm  | 589   | 4        | 2.10             | 3.17           | 1.03               | 2.14              | 0.60                      |
| SSp    | 515   | 5        | 3.33             | 5.36           | 1.42               | 3.53              | 0.57                      |
| CA3    | 494   | 10       | 2.47             | 4.52           | 0.88               | 2.63              | 0.60                      |
| LSr    | 475   | 4        | 2.06             | 2.66           | 1.09               | 2.07              | 0.53                      |
| VISI   | 434   | 3        | 2.55             | 3.82           | 1.19               | 2.75              | 0.57                      |
| ILA    | 411   | 3        | 2.83             | 3.52           | 1.57               | 2.52              | 0.63                      |
| LS     | 405   | 3        | 3.50             | 5.87           | 1.71               | 3.60              | 0.50                      |
| MD     | 397   | 3        | 4.61             | 4.51           | 3.47               | 4.89              | 0.53                      |
| SCm    | 356   | 7        | 5.13             | 8.11           | 2.24               | 5.06              | 0.27                      |
| VPM    | 353   | 4        | 3.25             | 3.92           | 1.66               | 3.35              | 0.43                      |
| VISa   | 340   | 5        | 1.82             | 2.64           | 0.86               | 1.82              | 0.60                      |
| ACB    | 337   | 3        | 1.29             | 2.24           | 0.50               | 1.51              | 0.63                      |
| BLA    | 317   | 2        | 1.47             | 2.85           | 0.70               | 1.51              | 0.63                      |
| COA    | 316   | 1        | 0.83             | 1.43           | 0.22               | 0.96              | 0.60                      |
| POST   | 312   | 4        | 3.51             | 4.36           | 2.07               | 3.45              | 0.50                      |
| SNr    | 308   | 4        | 8.20             | 14.07          | 3.25               | 8.45              | 0.27                      |
| VPL    | 301   | 4        | 6.52             | 8.65           | 2.74               | 6.92              | 0.43                      |
| LD     | 298   | 6        | 7.63             | 6.86           | 6.72               | 7.73              | 0.47                      |
| MG     | 295   | 2        | 5.80             | 4.44           | 5.27               | 5.88              | 0.43                      |

| Region | Units | Sessions | Spont. mean (Hz) | Spont. SD (Hz) | Spont. median (Hz) | Overall mean (Hz) | Waveform dur. median (ms) |
|--------|-------|----------|------------------|----------------|--------------------|-------------------|---------------------------|
| GPe    | 292   | 3        | 7.83             | 15.67          | 3.70               | 7.92              | 0.30                      |
| ZI     | 289   | 4        | 4.19             | 7.21           | 1.69               | 4.72              | 0.43                      |
| VISrl  | 286   | 2        | 0.91             | 1.19           | 0.55               | 0.76              | 0.63                      |
| OLF    | 263   | 3        | 1.56             | 2.98           | 0.80               | 1.41              | 0.60                      |
| APN    | 247   | 2        | 11.11            | 15.06          | 4.01               | 11.28             | 0.27                      |
| AUD    | 246   | 1        | 1.11             | 1.24           | 0.77               | 1.16              | 0.60                      |
| POL    | 215   | 3        | 4.36             | 7.59           | 2.00               | 4.54              | 0.43                      |
| CA     | 193   | 1        | 1.67             | 3.96           | 0.73               | 1.70              | 0.67                      |
| TT     | 181   | 4        | 2.86             | 6.12           | 0.83               | 2.75              | 0.57                      |
| SCsg   | 178   | 6        | 2.26             | 3.42           | 0.96               | 2.42              | 0.40                      |
| RT     | 173   | 2        | 3.96             | 4.51           | 2.20               | 4.08              | 0.27                      |
| SCs    | 157   | 4        | 2.14             | 3.31           | 0.90               | 2.69              | 0.27                      |
| PAG    | 141   | 3        | 5.18             | 6.63           | 2.63               | 4.74              | 0.23                      |
| ORBm   | 122   | 1        | 2.69             | 3.78           | 1.30               | 2.68              | 0.67                      |
| SSs    | 120   | 1        | 1.42             | 1.63           | 0.80               | 1.61              | 0.53                      |
| CA2    | 115   | 1        | 1.93             | 2.20           | 1.52               | 1.89              | 0.40                      |
| BMA    | 104   | 1        | 1.93             | 1.66           | 1.45               | 1.87              | 0.57                      |
| EP     | 91    | 1        | 1.77             | 2.08           | 1.05               | 1.54              | 0.43                      |
| VAL    | 83    | 1        | 11.75            | 11.21          | 9.07               | 11.85             | 0.43                      |
| LSc    | 72    | 1        | 2.06             | 3.30           | 0.45               | 1.96              | 0.50                      |
| PIR    | 67    | 1        | 4.54             | 5.23           | 2.39               | 4.37              | 0.57                      |
| IC     | 58    | 1        | 7.57             | 11.27          | 2.50               | 8.19              | 0.22                      |
| RN     | 58    | 1        | 16.72            | 20.62          | 6.92               | 18.11             | 0.27                      |
| SPF    | 53    | 2        | 9.67             | 15.15          | 2.40               | 10.03             | 0.37                      |
| EPd    | 52    | 1        | 1.23             | 1.77           | 0.78               | 1.04              | 0.63                      |
| SI     | 48    | 1        | 8.85             | 16.66          | 2.79               | 8.51              | 0.30                      |
| PT     | 45    | 1        | 3.45             | 4.61           | 1.46               | 3.75              | 0.50                      |
| NB     | 43    | 1        | 6.81             | 8.42           | 4.19               | 6.41              | 0.27                      |
| MEA    | 41    | 1        | 3.62             | 5.48           | 2.60               | 3.30              | 0.53                      |
| DP     | 39    | 1        | 3.83             | 6.24           | 1.33               | 3.20              | 0.60                      |
| CL     | 32    | 1        | 2.52             | 3.24           | 1.76               | 2.12              | 0.57                      |
| OT     | 26    | 1        | 1.27             | 1.65           | 0.41               | 1.02              | 0.62                      |
| LH     | 24    | 2        | 20.31            | 17.93          | 18.16              | 16.61             | 0.40                      |
| MS     | 23    | 1        | 17.26            | 19.24          | 8.40               | 17.28             | 0.33                      |

## 4. Unit-level data: ZI

| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-06 | 4    | 7424.0 | 3712.9 | 2768.7 | 198.6      | 0.37           | 476    | 0.14      | 0.16      | 0.14        |
| ZI     | Lederberg_2017-12-06 | 15   | 7424.0 | 3675.3 | 2782.4 | 244.6      | 0.43           | 1358   | 0.40      | 0.46      | 0.38        |
| ZI     | Lederberg_2017-12-06 | 57   | 7392.0 | 3750.5 | 2755.0 | 166.1      | 0.27           | 5487   | 1.62      | 1.63      | 1.82        |
| ZI     | Lederberg_2017-12-06 | 88   | 7408.0 | 3769.3 | 2748.2 | 139.0      | 0.43           | 699    | 0.21      | 0.26      | 0.22        |
| ZI     | Lederberg_2017-12-06 | 100  | 7376.0 | 3769.3 | 2748.2 | 139.1      | 0.30           | 16574  | 4.89      | 5.63      | 4.39        |
| ZI     | Lederberg_2017-12-06 | 112  | 7424.0 | 3675.3 | 2782.4 | 233.3      | 0.43           | 1789   | 0.53      | 0.81      | 0.44        |
| ZI     | Lederberg_2017-12-06 | 117  | 7424.0 | 3675.3 | 2782.4 | 232.1      | 0.30           | 33894  | 10.01     | 1.08      | 16.67       |
| ZI     | Lederberg_2017-12-06 | 145  | 7408.0 | 3844.5 | 2720.8 | 65.9       | 0.43           | 314    | 0.09      | 0.06      | 0.15        |
| ZI     | Lederberg_2017-12-06 | 177  | 7424.0 | 3712.9 | 2768.7 | 198.6      | 0.37           | 1591   | 0.47      | 0.50      | 0.50        |
| ZI     | Lederberg_2017-12-06 | 209  | 7376.0 | 3769.3 | 2748.2 | 117.1      | 0.17           | 72555  | 21.42     | 19.33     | 24.97       |
| ZI     | Lederberg_2017-12-06 | 227  | 7376.0 | 3882.0 | 2707.2 | 39.9       | 0.23           | 52370  | 15.46     | 23.26     | 11.77       |
| ZI     | Lederberg_2017-12-06 | 243  | 7408.0 | 3844.5 | 2720.8 | 55.6       | 0.37           | 60935  | 17.99     | 20.74     | 21.50       |
| ZI     | Lederberg_2017-12-06 | 247  | 7392.0 | 3712.9 | 2768.7 | 191.4      | 0.37           | 41005  | 12.11     | 8.43      | 14.90       |
| ZI     | Lederberg_2017-12-06 | 277  | 7392.0 | 3750.5 | 2755.0 | 165.8      | 0.23           | 15238  | 4.50      | 7.47      | 3.32        |
| ZI     | Lederberg_2017-12-06 | 286  | 7408.0 | 3694.1 | 2775.6 | 223.5      | 0.30           | 18351  | 5.42      | 3.84      | 5.93        |
| ZI     | Lederberg_2017-12-08 | 577  | 7860.7 | 3831.0 | 3486.5 | 1800.4     | 0.23           | 5016   | 1.40      | 0.10      | 2.47        |
| ZI     | Lederberg_2017-12-08 | 580  | 7889.0 | 4140.1 | 3564.3 | 1488.5     | 0.23           | 11729  | 3.27      | 6.17      | 0.37        |
| ZI     | Lederberg_2017-12-08 | 581  | 7889.0 | 4140.1 | 3564.3 | 1479.9     | 0.17           | 6104   | 1.70      | 0.34      | 3.38        |
| ZI     | Lederberg_2017-12-08 | 612  | 7874.9 | 3985.5 | 3525.4 | 1639.5     | 0.13           | 46379  | 12.95     | 14.20     | 11.61       |
| ZI     | Lederberg_2017-12-08 | 627  | 7889.0 | 4140.1 | 3564.3 | 1480.8     | 0.33           | 10352  | 2.89      | 6.17      | 0.20        |
| ZI     | Lederberg_2017-12-08 | 644  | 7880.2 | 3715.1 | 3446.4 | 1921.3     | 0.47           | 1372   | 0.38      | 0.18      | 0.13        |
| ZI     | Lederberg_2017-12-08 | 661  | 7890.8 | 3831.0 | 3475.5 | 1790.8     | 0.23           | 9834   | 2.75      | 2.65      | 2.50        |
| ZI     | Lederberg_2017-12-08 | 669  | 7860.7 | 3831.0 | 3486.5 | 1820.2     | 0.30           | 21128  | 5.90      | 5.34      | 1.41        |
| ZI     | Lederberg_2017-12-08 | 671  | 7925.3 | 4043.5 | 3523.6 | 1577.6     | 0.17           | 15311  | 4.27      | 2.83      | 5.47        |
| ZI     | Lederberg_2017-12-08 | 673  | 7878.4 | 4024.1 | 3535.1 | 1587.3     | 0.20           | 12662  | 3.54      | 6.52      | 0.92        |
| ZI     | Lederberg_2017-12-08 | 680  | 7860.7 | 3831.0 | 3486.5 | 1788.9     | 0.20           | 783    | 0.22      | 0.02      | 0.57        |
| ZI     | Lederberg_2017-12-08 | 683  | 7846.6 | 3676.4 | 3447.6 | 1949.2     | 0.60           | 28683  | 8.01      | 13.35     | 5.11        |
| ZI     | Lederberg_2017-12-08 | 706  | 7885.5 | 4101.4 | 3554.6 | 1519.7     | 0.27           | 1827   | 0.51      | 0.19      | 1.65        |
| ZI     | Lederberg_2017-12-08 | 707  | 7880.2 | 3715.1 | 3446.4 | 1922.9     | 0.47           | 640    | 0.18      | 0.35      | 0.00        |
| ZI     | Lederberg_2017-12-08 | 720  | 7860.7 | 3831.0 | 3486.5 | 1805.9     | 0.17           | 21544  | 6.01      | 0.00      | 12.80       |
| ZI     | Lederberg_2017-12-08 | 729  | 7905.8 | 4159.4 | 3563.7 | 1458.5     | 0.47           | 713    | 0.20      | 0.21      | 0.35        |
| ZI     | Lederberg_2017-12-08 | 733  | 7898.8 | 4082.1 | 3544.2 | 1543.2     | 0.30           | 27072  | 7.56      | 4.69      | 11.15       |
| ZI     | Lederberg_2017-12-08 | 742  | 7885.5 | 4101.4 | 3554.6 | 1534.8     | 0.40           | 19867  | 5.55      | 11.66     | 0.49        |
| ZI     | Lederberg_2017-12-08 | 745  | 7878.4 | 4024.1 | 3535.1 | 1587.2     | 0.20           | 19968  | 5.57      | 9.90      | 1.35        |
| ZI     | Lederberg_2017-12-08 | 746  | 7885.5 | 4101.4 | 3554.6 | 1530.2     | 0.27           | 2175   | 0.61      | 0.90      | 0.83        |
| ZI     | Lederberg_2017-12-08 | 750  | 7882.0 | 4062.8 | 3544.9 | 1521.7     | 0.30           | 1919   | 0.54      | 0.88      | 0.19        |
| ZI     | Lederberg_2017-12-08 | 756  | 7878.4 | 4024.1 | 3535.1 | 1590.0     | 0.20           | 20546  | 5.74      | 9.99      | 1.46        |
| ZI     | Lederberg_2017-12-08 | 799  | 7912.0 | 4062.8 | 3533.9 | 1570.8     | 0.27           | 62869  | 17.55     | 35.15     | 2.81        |
| ZI     | Lederberg_2017-12-08 | 803  | 7918.2 | 3966.2 | 3504.1 | 1658.3     | 0.27           | 21282  | 5.94      | 8.38      | 3.23        |
| ZI     | Lederberg_2017-12-08 | 804  | 7898.8 | 4082.1 | 3544.2 | 1510.5     | 0.27           | 1168   | 0.33      | 0.22      | 0.46        |

| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-08 | 807  | 7885.5 | 4101.4 | 3554.6 | 1520.7     | 0.33           | 6252   | 1.75      | 0.87      | 4.12        |
| ZI     | Lederberg_2017-12-08 | 816  | 7904.1 | 3811.6 | 3465.2 | 1831.9     | 0.23           | 3984   | 1.11      | 2.21      | 0.21        |
| ZI     | Lederberg_2017-12-08 | 822  | 7881.1 | 3888.9 | 3495.6 | 1742.2     | 0.17           | 32142  | 8.97      | 9.42      | 8.43        |
| ZI     | Lederberg_2017-12-08 | 857  | 7885.5 | 4101.4 | 3554.6 | 1522.2     | 0.40           | 12501  | 3.49      | 7.46      | 0.24        |
| ZI     | Lederberg_2017-12-08 | 868  | 7902.3 | 4120.7 | 3554.0 | 1502.8     | 0.30           | 2198   | 0.61      | 0.23      | 2.01        |
| ZI     | Lederberg_2017-12-08 | 870  | 7914.7 | 3927.6 | 3494.4 | 1708.9     | 0.20           | 15000  | 4.19      | 8.14      | 1.05        |
| ZI     | Lederberg_2017-12-08 | 913  | 7893.4 | 3695.7 | 3436.0 | 1935.0     | 0.30           | 13452  | 3.76      | 6.98      | 1.42        |
| ZI     | Lederberg_2017-12-08 | 914  | 7860.7 | 3831.0 | 3486.5 | 1813.3     | 0.20           | 6798   | 1.90      | 3.91      | 0.20        |
| ZI     | Lederberg_2017-12-08 | 934  | 7878.4 | 4024.1 | 3535.1 | 1587.2     | 0.23           | 8609   | 2.40      | 4.53      | 0.59        |
| ZI     | Lederberg_2017-12-08 | 954  | 7932.4 | 4120.7 | 3543.0 | 1517.8     | 0.17           | 48798  | 13.62     | 24.56     | 5.16        |
| ZI     | Lederberg_2017-12-08 | 990  | 7878.4 | 4024.1 | 3535.1 | 1590.7     | 0.23           | 23301  | 6.51      | 10.79     | 1.74        |
| ZI     | Lederberg_2017-12-08 | 999  | 7893.4 | 3695.7 | 3436.0 | 1946.2     | 0.30           | 17576  | 4.91      | 8.48      | 1.77        |
| ZI     | Lederberg_2017-12-08 | 1001 | 7898.8 | 4082.1 | 3544.2 | 2020.6     | 0.43           | 759    | 0.21      | 0.28      | 0.17        |
| ZI     | Lederberg_2017-12-08 | 1013 | 7900.5 | 3773.0 | 3455.5 | 1857.0     | 0.43           | 1438   | 0.40      | 0.39      | 0.38        |
| ZI     | Lederberg_2017-12-08 | 1020 | 7885.5 | 4101.4 | 3554.6 | 1512.0     | 0.20           | 21369  | 5.97      | 5.65      | 6.77        |
| ZI     | Lederberg_2017-12-08 | 1033 | 7885.5 | 4101.4 | 3554.6 | 1516.3     | 0.37           | 2048   | 0.57      | 1.07      | 0.18        |
| ZI     | Lederberg_2017-12-08 | 1053 | 7885.5 | 4101.4 | 3554.6 | 1517.2     | 0.20           | 24322  | 6.79      | 6.09      | 7.90        |
| ZI     | Lederberg_2017-12-08 | 1080 | 7918.2 | 3966.2 | 3504.1 | 1658.6     | 0.23           | 22433  | 6.26      | 8.73      | 3.38        |
| ZI     | Lederberg_2017-12-08 | 1084 | 7928.8 | 4082.1 | 3533.3 | 1545.0     | 0.20           | 16180  | 4.52      | 7.55      | 4.31        |
| ZI     | Lederberg_2017-12-08 | 1092 | 7897.0 | 3734.4 | 3445.7 | 1915.1     | 0.43           | 1743   | 0.49      | 0.05      | 0.32        |
| ZI     | Lederberg_2017-12-08 | 1104 | 7898.8 | 4082.1 | 3544.2 | 1564.2     | 0.30           | 2478   | 0.69      | 1.13      | 0.25        |
| ZI     | Lederberg_2017-12-08 | 1112 | 7904.1 | 3811.6 | 3465.2 | 1813.7     | 0.53           | 31319  | 8.74      | 13.76     | 1.76        |
| ZI     | Lederberg_2017-12-08 | 1117 | 7881.1 | 3888.9 | 3495.6 | 1731.7     | 0.20           | 78353  | 21.88     | 19.82     | 25.03       |
| ZI     | Lederberg_2017-12-08 | 1118 | 7928.8 | 4082.1 | 3533.3 | 1564.0     | 0.20           | 101145 | 28.24     | 30.50     | 27.09       |
| ZI     | Lederberg_2017-12-08 | 1119 | 7898.8 | 4082.1 | 3544.2 | 1548.6     | 0.30           | 91520  | 25.55     | 32.18     | 20.05       |
| ZI     | Lederberg_2017-12-08 | 1124 | 7864.3 | 3869.6 | 3496.2 | 1767.4     | 0.20           | 57763  | 16.13     | 15.08     | 15.51       |
| ZI     | Lederberg_2017-12-08 | 1125 | 7914.7 | 3927.6 | 3494.4 | 1707.2     | 0.17           | 78479  | 21.91     | 24.81     | 18.68       |
| ZI     | Lederberg_2017-12-08 | 1128 | 7898.8 | 4082.1 | 3544.2 | 1547.9     | 0.27           | 116173 | 32.43     | 42.62     | 27.29       |
| ZI     | Lederberg_2017-12-08 | 1129 | 7882.0 | 4062.8 | 3544.9 | 1551.7     | 0.27           | 133490 | 37.27     | 34.80     | 41.96       |
| ZI     | Lederberg_2017-12-08 | 1131 | 7908.5 | 4024.1 | 3524.2 | 1590.6     | 0.17           | 66962  | 18.70     | 22.16     | 15.82       |
| ZI     | Lederberg_2017-12-11 | 590  | 7905.4 | 4076.8 | 3483.9 | 1958.7     | 0.40           | 16948  | 4.85      | 2.16      | 6.31        |
| ZI     | Lederberg_2017-12-11 | 591  | 7920.1 | 3965.5 | 3430.8 | 2059.8     | 0.43           | 18541  | 5.30      | 10.93     | 2.64        |
| ZI     | Lederberg_2017-12-11 | 592  | 7854.1 | 3705.9 | 3343.1 | 2379.4     | 0.47           | 2061   | 0.59      | 0.73      | 0.57        |
| ZI     | Lederberg_2017-12-11 | 596  | 7869.5 | 3817.2 | 3385.4 | 2240.5     | 0.37           | 261    | 0.07      | 0.12      | 0.06        |
| ZI     | Lederberg_2017-12-11 | 601  | 7947.9 | 4058.3 | 3460.5 | 1970.2     | 0.43           | 4931   | 1.41      | 3.85      | 0.20        |
| ZI     | Lederberg_2017-12-11 | 604  | 7900.3 | 4039.7 | 3469.9 | 1996.6     | 0.60           | 1356   | 0.39      | 1.00      | 0.13        |
| ZI     | Lederberg_2017-12-11 | 613  | 7935.4 | 4076.8 | 3473.0 | 1958.4     | 0.57           | 5981   | 1.71      | 1.71      | 1.51        |
| ZI     | Lederberg_2017-12-11 | 616  | 7884.2 | 3705.9 | 3332.2 | 2369.3     | 0.43           | 7182   | 2.05      | 2.31      | 1.97        |
| ZI     | Lederberg_2017-12-11 | 617  | 7958.2 | 4132.4 | 3488.6 | 1908.3     | 0.53           | 560    | 0.16      | 0.27      | 0.19        |
| ZI     | Lederberg_2017-12-11 | 623  | 7923.0 | 4095.3 | 3485.5 | 1942.8     | 0.47           | 11749  | 3.36      | 1.48      | 4.32        |
| ZI     | Lederberg_2017-12-11 | 632  | 7915.6 | 4151.0 | 3512.1 | 1879.5     | 0.50           | 4776   | 1.37      | 0.93      | 1.29        |
| ZI     | Lederberg_2017-12-11 | 633  | 7902.5 | 3947.0 | 3429.2 | 2112.0     | 0.40           | 2540   | 0.73      | 1.50      | 0.21        |

| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-11 | 634  | 7901.8 | 3724.5 | 3333.8 | 2333.3     | 0.43           | 6486   | 1.85      | 1.25      | 2.25        |
| ZI     | Lederberg_2017-12-11 | 640  | 7953.0 | 4095.3 | 3474.6 | 1943.0     | 0.43           | 3846   | 1.10      | 0.92      | 1.08        |
| ZI     | Lederberg_2017-12-11 | 641  | 7854.1 | 3705.9 | 3343.1 | 2377.0     | 0.43           | 23568  | 6.74      | 6.49      | 7.09        |
| ZI     | Lederberg_2017-12-11 | 644  | 7906.9 | 3761.6 | 3347.8 | 2294.3     | 0.40           | 28354  | 8.11      | 7.85      | 8.72        |
| ZI     | Lederberg_2017-12-11 | 645  | 7869.5 | 3817.2 | 3385.4 | 2246.3     | 0.37           | 1492   | 0.43      | 0.19      | 0.60        |
| ZI     | Lederberg_2017-12-11 | 651  | 7884.2 | 3705.9 | 3332.2 | 2361.5     | 0.47           | 7054   | 2.02      | 1.38      | 2.59        |
| ZI     | Lederberg_2017-12-11 | 653  | 7864.4 | 3780.1 | 3371.3 | 2273.9     | 1.37           | 601    | 0.17      | 0.16      | 0.19        |
| ZI     | Lederberg_2017-12-11 | 664  | 7879.8 | 3891.4 | 3413.5 | 2178.8     | 0.60           | 2520   | 0.72      | 1.08      | 0.67        |
| ZI     | Lederberg_2017-12-11 | 667  | 7901.8 | 3724.5 | 3333.8 | 2351.9     | 0.47           | 6042   | 1.73      | 0.69      | 2.35        |
| ZI     | Lederberg_2017-12-11 | 678  | 7859.3 | 3743.0 | 3357.2 | 2312.4     | 0.47           | 14662  | 4.19      | 4.65      | 4.45        |
| ZI     | Lederberg_2017-12-11 | 679  | 7937.7 | 3984.1 | 3432.3 | 2055.2     | 0.47           | 13982  | 4.00      | 2.34      | 4.88        |
| ZI     | Lederberg_2017-12-11 | 684  | 7882.0 | 3798.6 | 3372.9 | 2261.7     | 0.50           | 7903   | 2.26      | 2.11      | 2.45        |
| ZI     | Lederberg_2017-12-11 | 686  | 7927.4 | 3909.9 | 3404.2 | 2152.6     | 0.53           | 6229   | 1.78      | 2.76      | 1.37        |
| ZI     | Lederberg_2017-12-11 | 701  | 7879.8 | 3891.4 | 3413.5 | 2149.0     | 0.47           | 282    | 0.08      | 0.15      | 0.05        |
| ZI     | Lederberg_2017-12-11 | 703  | 7900.3 | 4039.7 | 3469.9 | 2011.2     | 0.50           | 8112   | 2.32      | 3.33      | 1.96        |
| ZI     | Lederberg_2017-12-11 | 705  | 7937.7 | 3984.1 | 3432.3 | 2065.0     | 0.50           | 2258   | 0.65      | 0.82      | 0.58        |
| ZI     | Lederberg_2017-12-11 | 708  | 7942.8 | 4021.2 | 3446.4 | 2022.6     | 0.40           | 12507  | 3.58      | 5.75      | 2.40        |
| ZI     | Lederberg_2017-12-11 | 713  | 7859.3 | 3743.0 | 3357.2 | 2332.8     | 0.43           | 9126   | 2.61      | 2.95      | 2.55        |
| ZI     | Lederberg_2017-12-11 | 716  | 7942.8 | 4021.2 | 3446.4 | 1994.7     | 0.43           | 15009  | 4.29      | 6.04      | 3.39        |
| ZI     | Lederberg_2017-12-11 | 721  | 7906.9 | 3761.6 | 3347.8 | 2305.5     | 0.43           | 16992  | 4.86      | 1.87      | 6.37        |
| ZI     | Lederberg_2017-12-11 | 722  | 7890.0 | 3965.5 | 3441.7 | 2079.5     | 0.47           | 1567   | 0.45      | 1.10      | 0.12        |
| ZI     | Lederberg_2017-12-11 | 724  | 7869.5 | 3817.2 | 3385.4 | 2240.3     | 0.47           | 12036  | 3.44      | 4.52      | 3.23        |
| ZI     | Lederberg_2017-12-11 | 725  | 7912.0 | 3798.6 | 3361.9 | 2279.0     | 0.40           | 2281   | 0.65      | 0.55      | 0.76        |
| ZI     | Lederberg_2017-12-11 | 726  | 7859.3 | 3743.0 | 3357.2 | 2332.5     | 0.43           | 21133  | 6.04      | 7.77      | 5.63        |
| ZI     | Lederberg_2017-12-11 | 729  | 7947.9 | 4058.3 | 3460.5 | 1971.8     | 0.57           | 5054   | 1.45      | 2.36      | 1.05        |
| ZI     | Lederberg_2017-12-11 | 735  | 7953.0 | 4095.3 | 3474.6 | 1951.0     | 0.47           | 7720   | 2.21      | 1.37      | 2.89        |
| ZI     | Lederberg_2017-12-11 | 737  | 7879.8 | 3891.4 | 3413.5 | 2170.6     | 0.40           | 15631  | 4.47      | 6.68      | 3.41        |
| ZI     | Lederberg_2017-12-11 | 747  | 7854.1 | 3705.9 | 3343.1 | 2361.8     | 0.43           | 9433   | 2.70      | 1.50      | 3.53        |
| ZI     | Lederberg_2017-12-11 | 750  | 7947.9 | 4058.3 | 3460.5 | 1983.2     | 0.53           | 18839  | 5.39      | 5.50      | 5.29        |
| ZI     | Lederberg_2017-12-11 | 752  | 7927.4 | 3909.9 | 3404.2 | 2152.6     | 0.53           | 6952   | 1.99      | 1.19      | 2.88        |
| ZI     | Lederberg_2017-12-11 | 756  | 7915.6 | 4151.0 | 3512.1 | 1855.2     | 0.60           | 9988   | 2.86      | 2.95      | 2.62        |
| ZI     | Lederberg_2017-12-11 | 763  | 7910.5 | 4113.9 | 3498.0 | 1918.5     | 0.50           | 11161  | 3.19      | 3.87      | 2.75        |
| ZI     | Lederberg_2017-12-11 | 764  | 7879.8 | 3891.4 | 3413.5 | 2178.6     | 0.57           | 757    | 0.22      | 0.53      | 0.06        |
| ZI     | Lederberg_2017-12-11 | 765  | 7906.9 | 3761.6 | 3347.8 | 2297.4     | 0.47           | 1870   | 0.53      | 0.80      | 0.44        |
| ZI     | Lederberg_2017-12-11 | 767  | 7909.8 | 3891.4 | 3402.6 | 2442.7     | 0.53           | 149    | 0.04      | 0.07      | 0.03        |
| ZI     | Lederberg_2017-12-11 | 770  | 7930.3 | 4039.7 | 3458.9 | 1997.1     | 0.50           | 5805   | 1.66      | 2.16      | 1.26        |
| ZI     | Lederberg_2017-12-11 | 772  | 7869.5 | 3817.2 | 3385.4 | 2241.0     | 0.40           | 1048   | 0.30      | 0.20      | 0.44        |
| ZI     | Lederberg_2017-12-11 | 774  | 7879.8 | 3891.4 | 3413.5 | 2170.6     | 0.53           | 2289   | 0.65      | 1.41      | 0.30        |
| ZI     | Lederberg_2017-12-11 | 779  | 7889.3 | 3743.0 | 3346.3 | 2314.1     | 0.47           | 12270  | 3.51      | 8.92      | 0.98        |
| ZI     | Lederberg_2017-12-11 | 783  | 7909.8 | 3891.4 | 3402.6 | 2157.2     | 0.60           | 2585   | 0.74      | 1.97      | 0.11        |
| ZI     | Lederberg_2017-12-11 | 785  | 7895.1 | 4002.6 | 3455.8 | 2208.6     | 0.50           | 2078   | 0.59      | 0.92      | 0.44        |
| ZI     | Lederberg_2017-12-11 | 796  | 7895.1 | 4002.6 | 3455.8 | 2136.5     | 0.47           | 2744   | 0.78      | 1.50      | 0.44        |



| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-11 | 801  | 7874.6 | 3854.3 | 3399.5 | 2190.6     | 0.43           | 365    | 0.10      | 0.07      | 0.16        |
| ZI     | Lederberg_2017-12-11 | 803  | 7947.9 | 4058.3 | 3460.5 | 1972.4     | 0.50           | 5302   | 1.52      | 2.64      | 0.97        |
| ZI     | Lederberg_2017-12-11 | 804  | 7859.3 | 3743.0 | 3357.2 | 2328.7     | 0.47           | 777    | 0.22      | 0.21      | 0.26        |
| ZI     | Lederberg_2017-12-11 | 806  | 7869.5 | 3817.2 | 3385.4 | 2240.6     | 0.37           | 827    | 0.24      | 0.18      | 0.33        |
| ZI     | Lederberg_2017-12-11 | 807  | 7925.2 | 4002.6 | 3444.8 | 1932.7     | 0.50           | 3334   | 0.95      | 1.48      | 1.16        |
| ZI     | Lederberg_2017-12-11 | 819  | 7912.7 | 4021.2 | 3457.3 | 2028.1     | 0.47           | 15460  | 4.42      | 5.14      | 4.17        |
| ZI     | Lederberg_2017-12-11 | 820  | 7879.8 | 3891.4 | 3413.5 | 2159.8     | 0.47           | 4668   | 1.33      | 0.95      | 1.68        |
| ZI     | Lederberg_2017-12-11 | 822  | 7882.0 | 3798.6 | 3372.9 | 2259.3     | 0.50           | 7865   | 2.25      | 2.20      | 2.44        |
| ZI     | Lederberg_2017-12-11 | 823  | 7871.7 | 3724.5 | 3344.7 | 2333.2     | 0.40           | 14554  | 4.16      | 4.43      | 4.44        |
| ZI     | Lederberg_2017-12-11 | 827  | 7932.5 | 3947.0 | 3418.2 | 2100.0     | 1.30           | 50554  | 14.45     | 11.76     | 17.01       |
| ZI     | Lederberg_2017-12-11 | 830  | 7859.3 | 3743.0 | 3357.2 | 2316.5     | 0.40           | 21218  | 6.07      | 5.14      | 6.85        |
| ZI     | Lederberg_2017-12-11 | 843  | 7859.3 | 3743.0 | 3357.2 | 2332.1     | 0.50           | 1351   | 0.39      | 0.39      | 0.46        |
| ZI     | Lederberg_2017-12-11 | 849  | 7890.0 | 3965.5 | 3441.7 | 2693.5     | 0.50           | 323    | 0.09      | 0.17      | 0.05        |
| ZI     | Lederberg_2017-12-11 | 860  | 7879.8 | 3891.4 | 3413.5 | 2149.0     | 0.47           | 780    | 0.22      | 0.35      | 0.18        |
| ZI     | Lederberg_2017-12-11 | 861  | 7879.8 | 3891.4 | 3413.5 | 2175.3     | 0.57           | 510    | 0.15      | 0.04      | 0.23        |
| ZI     | Lederberg_2017-12-11 | 862  | 7884.9 | 3928.4 | 3427.6 | 2109.8     | 0.43           | 9532   | 2.73      | 4.51      | 1.67        |
| ZI     | Lederberg_2017-12-11 | 869  | 7912.0 | 3798.6 | 3361.9 | 2169.6     | 0.47           | 419    | 0.12      | 0.16      | 0.11        |
| ZI     | Lederberg_2017-12-11 | 879  | 7942.8 | 4021.2 | 3446.4 | 2000.0     | 0.53           | 11835  | 3.38      | 8.45      | 0.84        |
| ZI     | Lederberg_2017-12-11 | 886  | 7879.8 | 3891.4 | 3413.5 | 2174.7     | 0.60           | 4306   | 1.23      | 2.97      | 0.45        |
| ZI     | Lederberg_2017-12-11 | 890  | 7927.4 | 3909.9 | 3404.2 | 2121.4     | 0.47           | 3172   | 0.91      | 0.75      | 0.96        |
| ZI     | Lederberg_2017-12-11 | 892  | 7890.0 | 3965.5 | 3441.7 | 2067.5     | 0.43           | 13795  | 3.94      | 10.23     | 0.52        |
| ZI     | Lederberg_2017-12-11 | 896  | 7905.4 | 4076.8 | 3483.9 | 1968.5     | 0.43           | 9364   | 2.68      | 5.32      | 1.41        |
| ZI     | Lederberg_2017-12-11 | 908  | 7884.9 | 3928.4 | 3427.6 | 2131.4     | 0.57           | 14461  | 4.13      | 9.63      | 1.35        |
| ZI     | Lederberg_2017-12-11 | 912  | 7927.4 | 3909.9 | 3404.2 | 2127.8     | 0.47           | 6488   | 1.86      | 2.02      | 1.86        |
| ZI     | Lederberg_2017-12-11 | 914  | 7930.3 | 4039.7 | 3458.9 | 2002.3     | 0.33           | 8561   | 2.45      | 0.68      | 3.76        |
| ZI     | Lederberg_2017-12-11 | 917  | 7864.4 | 3780.1 | 3371.3 | 2288.2     | 0.40           | 8675   | 2.48      | 3.56      | 1.98        |
| ZI     | Lederberg_2017-12-11 | 920  | 7864.4 | 3780.1 | 3371.3 | 2273.4     | 0.40           | 8403   | 2.40      | 1.40      | 3.17        |
| ZI     | Lederberg_2017-12-11 | 921  | 7874.6 | 3854.3 | 3399.5 | 2190.6     | 0.43           | 679    | 0.19      | 0.19      | 0.21        |
| ZI     | Lederberg_2017-12-11 | 923  | 7953.0 | 4095.3 | 3474.6 | 2440.5     | 0.90           | 1926   | 0.55      | 0.77      | 0.46        |
| ZI     | Lederberg_2017-12-11 | 924  | 7947.9 | 4058.3 | 3460.5 | 1980.9     | 0.57           | 4949   | 1.42      | 1.96      | 1.03        |
| ZI     | Lederberg_2017-12-11 | 925  | 7907.6 | 3984.1 | 3443.3 | 2056.9     | 0.40           | 20050  | 5.73      | 3.66      | 6.55        |
| ZI     | Lederberg_2017-12-11 | 926  | 7927.4 | 3909.9 | 3404.2 | 2150.5     | 0.57           | 10533  | 3.01      | 4.73      | 2.17        |
| ZI     | Lederberg_2017-12-11 | 928  | 7859.3 | 3743.0 | 3357.2 | 2313.8     | 0.47           | 3923   | 1.12      | 0.12      | 1.92        |
| ZI     | Lederberg_2017-12-11 | 945  | 7942.8 | 4021.2 | 3446.4 | 1995.4     | 0.60           | 4446   | 1.27      | 2.97      | 0.41        |
| ZI     | Lederberg_2017-12-11 | 947  | 7859.3 | 3743.0 | 3357.2 | 2337.7     | 0.43           | 505    | 0.14      | 0.11      | 0.20        |
| ZI     | Lederberg_2017-12-11 | 948  | 7905.4 | 4076.8 | 3483.9 | 1956.1     | 0.50           | 6985   | 2.00      | 5.03      | 0.54        |
| ZI     | Lederberg_2017-12-11 | 953  | 7869.5 | 3817.2 | 3385.4 | 2240.5     | 0.37           | 412    | 0.12      | 0.09      | 0.17        |
| ZI     | Lederberg_2017-12-11 | 954  | 7923.0 | 4095.3 | 3485.5 | 1938.0     | 0.60           | 12086  | 3.46      | 9.39      | 0.56        |
| ZI     | Lederberg_2017-12-11 | 956  | 7884.9 | 3928.4 | 3427.6 | 2105.4     | 0.47           | 11581  | 3.31      | 8.34      | 0.70        |
| ZI     | Lederberg_2017-12-11 | 957  | 7864.4 | 3780.1 | 3371.3 | 2283.4     | 0.43           | 28691  | 8.20      | 9.10      | 7.97        |
| ZI     | Lederberg_2017-12-11 | 963  | 7901.8 | 3724.5 | 3333.8 | 2324.0     | 0.43           | 2148   | 0.61      | 0.89      | 0.51        |
| ZI     | Lederberg_2017-12-11 | 974  | 7901.8 | 3724.5 | 3333.8 | 2350.4     | 0.43           | 2637   | 0.75      | 0.40      | 1.02        |

| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (µm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-11 | 975  | 7884.2 | 3705.9 | 3332.2 | 2369.8     | 0.43           | 15704  | 4.49      | 4.57      | 4.75        |
| ZI     | Lederberg_2017-12-11 | 977  | 7937.7 | 3984.1 | 3432.3 | 2058.3     | 0.57           | 9400   | 2.69      | 2.56      | 2.69        |
| ZI     | Lederberg_2017-12-11 | 983  | 7869.5 | 3817.2 | 3385.4 | 2250.7     | 0.63           | 5339   | 1.53      | 4.26      | 0.17        |
| ZI     | Lederberg_2017-12-11 | 984  | 7912.7 | 4021.2 | 3457.3 | 2030.6     | 0.50           | 6673   | 1.91      | 3.41      | 1.25        |
| ZI     | Lederberg_2017-12-11 | 987  | 7890.0 | 3965.5 | 3441.7 | 2092.4     | 0.40           | 7982   | 2.28      | 1.10      | 2.47        |
| ZI     | Lederberg_2017-12-11 | 988  | 7874.6 | 3854.3 | 3399.5 | 2206.4     | 0.47           | 2235   | 0.64      | 0.65      | 0.77        |
| ZI     | Lederberg_2017-12-11 | 989  | 7902.5 | 3947.0 | 3429.2 | 2112.0     | 0.40           | 4185   | 1.20      | 1.91      | 0.84        |
| ZI     | Lederberg_2017-12-11 | 993  | 7927.4 | 3909.9 | 3404.2 | 2137.8     | 0.53           | 2352   | 0.67      | 1.47      | 0.30        |
| ZI     | Lederberg_2017-12-11 | 997  | 7906.9 | 3761.6 | 3347.8 | 2302.6     | 0.47           | 8314   | 2.38      | 5.24      | 0.91        |
| ZI     | Lederberg_2017-12-11 | 1001 | 7879.8 | 3891.4 | 3413.5 | 2169.8     | 0.57           | 2805   | 0.80      | 1.74      | 0.37        |
| ZI     | Lederberg_2017-12-11 | 1007 | 7912.0 | 3798.6 | 3361.9 | 2258.9     | 0.40           | 2640   | 0.75      | 0.77      | 0.69        |
| ZI     | Lederberg_2017-12-11 | 1010 | 7859.3 | 3743.0 | 3357.2 | 2330.9     | 0.47           | 6348   | 1.82      | 2.05      | 1.79        |
| ZI     | Lederberg_2017-12-11 | 1020 | 7942.8 | 4021.2 | 3446.4 | 2025.2     | 0.40           | 1606   | 0.46      | 0.19      | 0.59        |
| ZI     | Lederberg_2017-12-11 | 1026 | 7905.4 | 4076.8 | 3483.9 | 1979.5     | 0.50           | 8842   | 2.53      | 3.80      | 1.89        |
| ZI     | Lederberg_2017-12-11 | 1030 | 7905.4 | 4076.8 | 3483.9 | 1944.5     | 0.43           | 16180  | 4.63      | 5.24      | 4.60        |
| ZI     | Lederberg_2017-12-11 | 1038 | 7910.5 | 4113.9 | 3498.0 | 1899.8     | 0.57           | 2372   | 0.68      | 1.81      | 0.10        |
| ZI     | Lederberg_2017-12-11 | 1042 | 7882.0 | 3798.6 | 3372.9 | 2262.7     | 0.47           | 6561   | 1.88      | 1.94      | 1.97        |
| ZI     | Lederberg_2017-12-11 | 1046 | 7859.3 | 3743.0 | 3357.2 | 2326.8     | 0.47           | 3159   | 0.90      | 1.06      | 0.94        |
| ZI     | Lederberg_2017-12-11 | 1048 | 7869.5 | 3817.2 | 3385.4 | 2257.6     | 0.40           | 11273  | 3.22      | 3.63      | 3.21        |
| ZI     | Lederberg_2017-12-11 | 1060 | 7859.3 | 3743.0 | 3357.2 | 2373.2     | 0.47           | 1383   | 0.40      | 0.40      | 0.41        |
| ZI     | Lederberg_2017-12-11 | 1061 | 7906.9 | 3761.6 | 3347.8 | 2392.5     | 0.53           | 541    | 0.15      | 0.26      | 0.11        |
| ZI     | Lederberg_2017-12-11 | 1072 | 7859.3 | 3743.0 | 3357.2 | 2328.7     | 0.47           | 913    | 0.26      | 0.34      | 0.26        |
| ZI     | Lederberg_2017-12-11 | 1076 | 7955.9 | 4225.1 | 3529.3 | 1798.9     | 0.97           | 10120  | 2.89      | 1.31      | 3.16        |
| ZI     | Lederberg_2017-12-11 | 1080 | 7968.4 | 4206.6 | 3516.8 | 1820.0     | 1.40           | 48075  | 13.75     | 37.68     | 1.99        |
| ZI     | Lederberg_2017-12-11 | 1087 | 7879.8 | 3891.4 | 3413.5 | 2439.4     | 0.50           | 244    | 0.07      | 0.07      | 0.07        |
| ZI     | Lederberg_2017-12-11 | 1088 | 7912.7 | 4021.2 | 3457.3 | 2027.6     | 0.43           | 17713  | 5.06      | 4.74      | 5.39        |
| ZI     | Lederberg_2017-12-11 | 1094 | 7890.0 | 3965.5 | 3441.7 | 2094.0     | 0.47           | 12159  | 3.48      | 7.51      | 1.12        |
| ZI     | Lederberg_2017-12-11 | 1100 | 7905.4 | 4076.8 | 3483.9 | 1949.4     | 0.53           | 5312   | 1.52      | 3.44      | 0.56        |
| ZI     | Lederberg_2017-12-11 | 1117 | 7874.6 | 3854.3 | 3399.5 | 2189.7     | 0.53           | 2983   | 0.85      | 2.26      | 0.20        |
| ZI     | Lederberg_2017-12-11 | 1120 | 7854.1 | 3705.9 | 3343.1 | 2378.1     | 0.50           | 2872   | 0.82      | 0.80      | 0.84        |
| ZI     | Lederberg_2017-12-11 | 1125 | 7879.8 | 3891.4 | 3413.5 | 2175.2     | 0.43           | 2997   | 0.86      | 0.96      | 0.78        |
| ZI     | Lederberg_2017-12-11 | 1127 | 7917.2 | 3835.7 | 3376.0 | 2234.7     | 0.47           | 10225  | 2.92      | 4.59      | 2.13        |
| ZI     | Lederberg_2017-12-11 | 1135 | 7900.3 | 4039.7 | 3469.9 | 1984.4     | 0.53           | 15651  | 4.48      | 8.39      | 2.66        |
| ZI     | Lederberg_2017-12-11 | 1136 | 7879.8 | 3891.4 | 3413.5 | 2161.7     | 0.53           | 6962   | 1.99      | 1.08      | 2.63        |
| ZI     | Lederberg_2017-12-11 | 1139 | 7927.4 | 3909.9 | 3404.2 | 2145.2     | 0.57           | 13322  | 3.81      | 5.45      | 2.86        |
| ZI     | Lederberg_2017-12-11 | 1142 | 7905.4 | 4076.8 | 3483.9 | 2350.4     | 0.47           | 579    | 0.17      | 0.11      | 0.18        |
| ZI     | Lederberg_2017-12-11 | 1150 | 7874.6 | 3854.3 | 3399.5 | 2211.3     | 0.50           | 5474   | 1.57      | 1.98      | 1.44        |
| ZI     | Lederberg_2017-12-11 | 1156 | 7906.9 | 3761.6 | 3347.8 | 2392.5     | 0.53           | 251    | 0.07      | 0.09      | 0.07        |
| ZI     | Lederberg_2017-12-11 | 1158 | 7932.5 | 3947.0 | 3418.2 | 2100.0     | 1.37           | 38139  | 10.91     | 7.27      | 10.09       |
| ZI     | Lederberg_2017-12-11 | 1168 | 7874.6 | 3854.3 | 3399.5 | 2310.2     | 0.47           | 1331   | 0.38      | 0.54      | 0.32        |
| ZI     | Lederberg_2017-12-11 | 1178 | 7869.5 | 3817.2 | 3385.4 | 2048.2     | 0.50           | 1623   | 0.46      | 0.69      | 0.43        |
| ZI     | Lederberg_2017-12-11 | 1186 | 7905.4 | 4076.8 | 3483.9 | 1962.2     | 0.47           | 11090  | 3.17      | 4.09      | 2.73        |

| Region | Session              | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|----------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Lederberg_2017-12-11 | 1189 | 7920.8 | 4188.1 | 3526.2 | 1850.8     | 0.97           | 47670  | 13.63     | 12.58     | 11.47       |
| ZI     | Lederberg_2017-12-11 | 1192 | 7909.8 | 3891.4 | 3402.6 | 2157.2     | 0.60           | 1593   | 0.46      | 0.85      | 0.26        |
| ZI     | Lederberg_2017-12-11 | 1206 | 7899.6 | 3817.2 | 3374.4 | 2251.0     | 0.50           | 9388   | 2.68      | 4.55      | 1.69        |
| ZI     | Lederberg_2017-12-11 | 1213 | 7869.5 | 3817.2 | 3385.4 | 2048.2     | 0.50           | 3179   | 0.91      | 1.09      | 0.92        |
| ZI     | Lederberg_2017-12-11 | 1214 | 7884.2 | 3705.9 | 3332.2 | 2353.7     | 0.40           | 5766   | 1.65      | 1.31      | 2.01        |
| ZI     | Lederberg_2017-12-11 | 1217 | 7900.3 | 4039.7 | 3469.9 | 1990.5     | 1.20           | 3485   | 1.00      | 0.25      | 0.59        |
| ZI     | Lederberg_2017-12-11 | 1228 | 7854.1 | 3705.9 | 3343.1 | 2379.5     | 0.50           | 6551   | 1.87      | 0.90      | 2.53        |
| ZI     | Lederberg_2017-12-11 | 1230 | 7947.9 | 4058.3 | 3460.5 | 1985.6     | 0.57           | 1323   | 0.38      | 0.70      | 0.25        |
| ZI     | Lederberg_2017-12-11 | 1242 | 7912.7 | 4021.2 | 3457.3 | 2018.6     | 0.40           | 4070   | 1.16      | 2.27      | 0.64        |
| ZI     | Lederberg_2017-12-11 | 1243 | 7912.0 | 3798.6 | 3361.9 | 2276.3     | 0.40           | 2777   | 0.79      | 0.57      | 1.06        |
| ZI     | Lederberg_2017-12-11 | 1244 | 7917.9 | 4058.3 | 3471.4 | 1984.4     | 0.47           | 15803  | 4.52      | 7.67      | 2.60        |
| ZI     | Lederberg_2017-12-11 | 1248 | 7915.6 | 4151.0 | 3512.1 | 1851.4     | 0.57           | 10234  | 2.93      | 2.91      | 2.70        |
| ZI     | Lederberg_2017-12-11 | 1250 | 7925.2 | 4002.6 | 3444.8 | 2044.1     | 0.47           | 5035   | 1.44      | 1.99      | 1.11        |
| ZI     | Lederberg_2017-12-11 | 1253 | 7879.8 | 3891.4 | 3413.5 | 2175.3     | 0.67           | 267    | 0.08      | 0.01      | 0.14        |
| ZI     | Lederberg_2017-12-11 | 1257 | 7917.2 | 3835.7 | 3376.0 | 2226.6     | 0.40           | 12896  | 3.69      | 4.55      | 3.25        |
| ZI     | Lederberg_2017-12-11 | 1262 | 7854.1 | 3705.9 | 3343.1 | 2375.8     | 0.47           | 2834   | 0.81      | 1.57      | 0.47        |
| ZI     | Lederberg_2017-12-11 | 1264 | 7917.2 | 3835.7 | 3376.0 | 2235.6     | 0.47           | 15539  | 4.44      | 6.84      | 3.29        |
| ZI     | Lederberg_2017-12-11 | 1272 | 7879.8 | 3891.4 | 3413.5 | 2175.3     | 0.67           | 340    | 0.10      | 0.02      | 0.15        |
| ZI     | Lederberg_2017-12-11 | 1274 | 7895.1 | 4002.6 | 3455.8 | 2042.3     | 0.47           | 6134   | 1.75      | 2.12      | 1.54        |
| ZI     | Lederberg_2017-12-11 | 1276 | 7906.9 | 3761.6 | 3347.8 | 2302.1     | 0.43           | 868    | 0.25      | 0.35      | 0.22        |
| ZI     | Lederberg_2017-12-11 | 1283 | 7864.4 | 3780.1 | 3371.3 | 2284.9     | 0.43           | 24922  | 7.13      | 8.00      | 6.98        |
| ZI     | Lederberg_2017-12-11 | 1284 | 7869.5 | 3817.2 | 3385.4 | 2240.6     | 0.37           | 301    | 0.09      | 0.18      | 0.05        |
| ZI     | Lederberg_2017-12-11 | 1285 | 7906.9 | 3761.6 | 3347.8 | 2322.9     | 0.40           | 3011   | 0.86      | 0.62      | 1.04        |
| ZI     | Lederberg_2017-12-11 | 1294 | 7869.5 | 3817.2 | 3385.4 | 2226.2     | 0.43           | 5526   | 1.58      | 1.89      | 1.68        |
| ZI     | Lederberg_2017-12-11 | 1295 | 7854.1 | 3705.9 | 3343.1 | 2376.0     | 0.53           | 1436   | 0.41      | 0.27      | 0.47        |
| ZI     | Lederberg_2017-12-11 | 1296 | 7871.7 | 3724.5 | 3344.7 | 2349.6     | 0.47           | 5213   | 1.49      | 1.16      | 1.95        |
| ZI     | Lederberg_2017-12-11 | 1302 | 7869.5 | 3817.2 | 3385.4 | 2241.0     | 0.40           | 677    | 0.19      | 0.33      | 0.14        |
| ZI     | Lederberg_2017-12-11 | 1303 | 7900.3 | 4039.7 | 3469.9 | 2015.1     | 0.50           | 7727   | 2.21      | 4.10      | 1.25        |
| ZI     | Lederberg_2017-12-11 | 1306 | 7874.6 | 3854.3 | 3399.5 | 2219.5     | 0.50           | 2147   | 0.61      | 0.72      | 0.66        |
| ZI     | Lederberg_2017-12-11 | 1314 | 7854.1 | 3705.9 | 3343.1 | 2379.0     | 0.50           | 2035   | 0.58      | 1.10      | 0.34        |
| ZI     | Lederberg_2017-12-11 | 1316 | 7859.3 | 3743.0 | 3357.2 | 2388.6     | 0.47           | 2844   | 0.81      | 0.94      | 0.76        |
| ZI     | Lederberg_2017-12-11 | 1318 | 7859.3 | 3743.0 | 3357.2 | 2373.2     | 0.47           | 630    | 0.18      | 0.12      | 0.24        |
| ZI     | Lederberg_2017-12-11 | 1319 | 7937.7 | 3984.1 | 3432.3 | 2070.0     | 0.47           | 17468  | 4.99      | 3.76      | 5.88        |
| ZI     | Lederberg_2017-12-11 | 1326 | 7942.8 | 4021.2 | 3446.4 | 1985.6     | 0.17           | 38027  | 10.87     | 5.17      | 13.66       |
| ZI     | Lederberg_2017-12-11 | 1333 | 7938.4 | 4206.6 | 3527.7 | 1816.4     | 0.43           | 6614   | 1.89      | 0.48      | 1.82        |
| ZI     | Lederberg_2017-12-11 | 1334 | 7935.4 | 4076.8 | 3473.0 | 1956.6     | 0.37           | 15749  | 4.50      | 3.52      | 5.09        |
| ZI     | Lederberg_2017-12-11 | 1337 | 7923.0 | 4095.3 | 3485.5 | 1939.1     | 0.43           | 4820   | 1.38      | 0.49      | 0.46        |
| ZI     | Richards_2017-11-01  | 478  | 7950.1 | 4483.9 | 3291.7 | 1312.9     | 0.20           | 6303   | 2.28      | 3.38      | 1.97        |
| ZI     | Richards_2017-11-01  | 526  | 7950.1 | 4483.9 | 3291.7 | 1303.9     | 0.20           | 2901   | 1.05      | 0.54      | 0.91        |
| ZI     | Richards_2017-11-01  | 543  | 7900.5 | 3943.0 | 3155.5 | 1866.2     | 0.37           | 8142   | 2.95      | 5.14      | 2.27        |
| ZI     | Richards_2017-11-01  | 547  | 7914.7 | 4097.6 | 3194.4 | 1700.0     | 0.20           | 25992  | 9.41      | 0.05      | 13.86       |
| ZI     | Richards_2017-11-01  | 599  | 7867.8 | 4078.2 | 3205.9 | 1725.6     | 0.17           | 1335   | 0.48      | 0.28      | 0.32        |

| Region | Session             | Unit | CCF AP | CCF DV | CCF LR | Depth (μm) | Wave dur. (ms) | Spikes | Mean (Hz) | Task (Hz) | Spont. (Hz) |
|--------|---------------------|------|--------|--------|--------|------------|----------------|--------|-----------|-----------|-------------|
| ZI     | Richards_2017-11-01 | 613  | 7892.6 | 4348.7 | 3274.0 | 1426.7     | 0.20           | 23356  | 8.46      | 8.10      | 9.06        |
| ZI     | Richards_2017-11-01 | 614  | 7867.8 | 4078.2 | 3205.9 | 1718.0     | 0.17           | 7564   | 2.74      | 1.45      | 3.23        |
| ZI     | Richards_2017-11-01 | 630  | 7920.9 | 4657.8 | 3351.9 | 1105.4     | 0.23           | 6371   | 2.31      | 1.15      | 2.75        |
| ZI     | Richards_2017-11-01 | 638  | 7939.5 | 4368.0 | 3262.5 | 1407.3     | 0.20           | 2318   | 0.84      | 1.00      | 0.78        |
| ZI     | Richards_2017-11-01 | 641  | 7896.1 | 4387.3 | 3283.8 | 1395.1     | 0.20           | 6921   | 2.51      | 4.22      | 1.81        |
| ZI     | Richards_2017-11-01 | 650  | 7913.8 | 4580.5 | 3332.4 | 1209.6     | 0.30           | 21261  | 7.70      | 3.06      | 8.88        |
| ZI     | Richards_2017-11-01 | 670  | 7925.3 | 4213.5 | 3223.6 | 1575.5     | 0.20           | 4431   | 1.60      | 5.89      | 0.36        |
| ZI     | Richards_2017-11-01 | 679  | 7953.6 | 4522.6 | 3301.4 | 1266.9     | 0.20           | 16541  | 5.99      | 2.90      | 7.91        |
| ZI     | Richards_2017-11-01 | 680  | 7899.7 | 4426.0 | 3293.5 | 1372.2     | 0.17           | 19851  | 7.19      | 6.80      | 8.24        |
| ZI     | Richards_2017-11-01 | 681  | 7871.3 | 4116.9 | 3215.7 | 1659.9     | 0.13           | 1572   | 0.57      | 0.29      | 0.39        |
| ZI     | Richards_2017-11-01 | 716  | 7890.8 | 4001.0 | 3175.5 | 1810.5     | 0.23           | 10947  | 3.97      | 9.26      | 2.24        |
| ZI     | Richards_2017-11-01 | 754  | 7939.5 | 4368.0 | 3262.5 | 1408.7     | 0.20           | 896    | 0.32      | 0.05      | 0.34        |
| ZI     | Richards_2017-11-01 | 769  | 7913.8 | 4580.5 | 3332.4 | 1194.0     | 0.17           | 12710  | 4.60      | 4.20      | 4.57        |
| ZI     | Richards_2017-11-01 | 786  | 7967.8 | 4677.1 | 3340.3 | 1101.3     | 0.17           | 81368  | 29.47     | 18.39     | 34.30       |
| ZI     | Richards_2017-11-01 | 787  | 7964.2 | 4638.5 | 3330.6 | 1156.5     | 0.20           | 150942 | 54.67     | 47.98     | 62.42       |
| ZI     | Richards_2017-11-01 | 788  | 7960.7 | 4599.8 | 3320.9 | 1182.9     | 0.33           | 31883  | 11.55     | 10.05     | 13.15       |
| ZI     | Richards_2017-11-01 | 789  | 7913.8 | 4580.5 | 3332.4 | 1219.7     | 0.23           | 42066  | 15.24     | 11.43     | 16.26       |
| ZI     | Richards_2017-11-01 | 790  | 7910.3 | 4541.9 | 3322.7 | 1232.9     | 0.23           | 99105  | 35.90     | 32.36     | 34.59       |
| ZI     | Richards_2017-11-01 | 791  | 7906.7 | 4503.2 | 3313.0 | 1292.9     | 0.17           | 30360  | 11.00     | 13.00     | 10.95       |
| ZI     | Richards_2017-11-01 | 792  | 7916.5 | 4445.3 | 3292.9 | 1335.4     | 0.37           | 76077  | 27.56     | 35.75     | 22.22       |
| ZI     | Richards_2017-11-01 | 793  | 7946.5 | 4445.3 | 3281.9 | 1350.0     | 0.20           | 88878  | 32.19     | 62.82     | 22.54       |
| ZI     | Richards_2017-11-01 | 794  | 7922.7 | 4348.7 | 3263.1 | 1438.6     | 0.27           | 64749  | 23.45     | 44.65     | 15.93       |
| ZI     | Richards_2017-11-01 | 795  | 7939.5 | 4368.0 | 3262.5 | 1429.9     | 0.17           | 30931  | 11.20     | 24.58     | 5.23        |
| ZI     | Richards_2017-11-01 | 796  | 7935.9 | 4329.4 | 3252.8 | 1472.4     | 0.10           | 10625  | 3.85      | 5.96      | 3.16        |
| ZI     | Richards_2017-11-01 | 797  | 7935.9 | 4329.4 | 3252.8 | 1461.2     | 0.23           | 21297  | 7.71      | 22.01     | 2.63        |
| ZI     | Richards_2017-11-01 | 798  | 7889.0 | 4310.1 | 3264.3 | 1477.7     | 0.20           | 41416  | 15.00     | 18.88     | 14.96       |
| ZI     | Richards_2017-11-01 | 799  | 7885.5 | 4271.4 | 3254.6 | 1513.7     | 1.40           | 10016  | 3.63      | 2.44      | 4.56        |
| ZI     | Richards_2017-11-01 | 800  | 7885.5 | 4271.4 | 3254.6 | 1508.0     | 0.17           | 25315  | 9.17      | 14.64     | 7.57        |
| ZI     | Richards_2017-11-01 | 801  | 7925.3 | 4213.5 | 3223.6 | 1580.2     | 0.17           | 15077  | 5.46      | 6.78      | 4.62        |
| ZI     | Richards_2017-11-01 | 802  | 7925.3 | 4213.5 | 3223.6 | 1595.5     | 0.17           | 14940  | 5.41      | 12.44     | 3.03        |
| ZI     | Richards_2017-11-01 | 803  | 7878.4 | 4194.1 | 3235.1 | 1600.8     | 0.17           | 3788   | 1.37      | 1.79      | 1.02        |
| ZI     | Richards_2017-11-01 | 804  | 7921.8 | 4174.8 | 3213.8 | 1623.5     | 0.20           | 47237  | 17.11     | 28.57     | 11.00       |
| ZI     | Richards_2017-11-01 | 805  | 7874.9 | 4155.5 | 3225.4 | 1642.1     | 0.17           | 35318  | 12.79     | 11.36     | 12.64       |
| ZI     | Richards_2017-11-01 | 806  | 7871.3 | 4116.9 | 3215.7 | 1699.6     | 0.17           | 38405  | 13.91     | 14.83     | 15.76       |
| ZI     | Richards_2017-11-01 | 807  | 7914.7 | 4097.6 | 3194.4 | 1706.0     | 0.13           | 88816  | 32.17     | 32.13     | 29.45       |
| ZI     | Richards_2017-11-01 | 808  | 7867.8 | 4078.2 | 3205.9 | 1731.6     | 0.17           | 23135  | 8.38      | 13.56     | 7.25        |
| ZI     | Richards_2017-11-01 | 809  | 7911.1 | 4058.9 | 3184.7 | 1758.9     | 0.20           | 26974  | 9.77      | 12.04     | 8.36        |
| ZI     | Richards_2017-11-01 | 810  | 7853.6 | 3923.7 | 3167.0 | 1869.8     | 0.50           | 18782  | 6.80      | 7.67      | 4.97        |
| ZI     | Richards_2017-11-01 | 811  | 7883.7 | 3923.7 | 3156.1 | 1873.3     | 0.63           | 25557  | 9.26      | 16.28     | 4.03        |

## 5. Sessions

| Session              | Sex | Age      | Genotype            | Duration (s) | Trials | Units | Good | Regions                                                                |
|----------------------|-----|----------|---------------------|--------------|--------|-------|------|------------------------------------------------------------------------|
| Cori_2016-12-14      | F   | 77 days  | tetO-G6s x CaMK-tTA | 2703         | 214    | 1085  | 734  | ACA, CA3, DG, LS, MOs, SUB, VISp, root                                 |
| Cori_2016-12-17      | F   | 84 days  | tetO-G6s x CaMK-tTA | 3147         | 251    | 1146  | 1070 | CA1, POST, VISI, VISpm, root                                           |
| Cori_2016-12-18      | F   | 84 days  | tetO-G6s x CaMK-tTA | 3834         | 228    | 1089  | 619  | CA1, DG, LP, MG, MRN, NB, POST, SPF, VISam, VISp, root                 |
| Forssmann_2017-11-01 | M   | 182 days | C57BL/6J            | 3115         | 249    | 2073  | 1769 | ACA, CA1, DG, LGd, LSr, MOs, SUB, TH, VISa, VISp, VPL                  |
| Forssmann_2017-11-02 | M   | 182 days | C57BL/6J            | 3044         | 254    | 1569  | 1077 | ACA, CA1, DG, MOs, OLF, ORB, PL, SUB, VISa, root                       |
| Forssmann_2017-11-04 | M   | 189 days | C57BL/6J            | 3472         | 290    | 1435  | 1169 | AUD, CA1, SSp, TH, root                                                |
| Forssmann_2017-11-05 | M   | 189 days | C57BL/6J            | 3123         | 252    | 1029  | 584  | CA3, CP, EPd, LD, PIR, SSp, VPL, root                                  |
| Hench_2017-06-15     | M   | 217 days | tetO-G6s x CaMK-tTA | 3260         | 250    | 1707  | 1157 | CA1, CA3, DG, ILA, LD, LP, LSr, MOs, PL, PO, SUB, TT, VISa, VISp, root |
| Hench_2017-06-16     | M   | 224 days | tetO-G6s x CaMK-tTA | 3915         | 372    | 1364  | 788  | CA1, CA3, LD, LSr, ORBm, PL, TH, TT, VISam, VISI, VPL, root            |
| Hench_2017-06-17     | M   | 224 days | tetO-G6s x CaMK-tTA | 4083         | 447    | 1591  | 1172 | CA1, DG, GPe, MB, MRN, POL, POST, SCm, SCsg, VISI, VISp, VISrl, root   |
| Hench_2017-06-18     | M   | 224 days | tetO-G6s x CaMK-tTA | 3720         | 342    | 1409  | 857  | CP, LSc, LSr, MOp, PT, root                                            |
| Lederberg_2017-12-05 | F   | 105 days | Vglut1-Cre          | 3446         | 340    | 1219  | 698  | ACA, CA1, DG, LGd, LH, MD, MOs, PL, SUB, VISam, VISp, root             |
| Lederberg_2017-12-06 | F   | 105 days | Vglut1-Cre          | 3387         | 300    | 1262  | 983  | ACA, CA1, DG, LGd, MB, MOs, MRN, MS, PL, RN, SCm, SCs, VISam, ZI, root |
| Lederberg_2017-12-07 | F   | 105 days | Vglut1-Cre          | 3195         | 268    | 1228  | 756  | CA1, MOs, MRN, ORB, PAG, RSP, SCm, SCs, VISp, root                     |
| Lederberg_2017-12-08 | F   | 105 days | Vglut1-Cre          | 3582         | 404    | 1134  | 743  | BLA, CA3, GPe, LGd, MB, VPM, ZI, root                                  |
| Lederberg_2017-12-09 | F   | 105 days | Vglut1-Cre          | 3128         | 280    | 977   | 474  | CA3, LGd, MB, SSp, SSs, TH, root                                       |
| Lederberg_2017-12-10 | F   | 105 days | Vglut1-Cre          | 2894         | 224    | 727   | 565  | LD, MEA, RT, VPL, VPM, root                                            |
| Lederberg_2017-12-11 | F   | 105 days | Vglut1-Cre          | 3497         | 316    | 1350  | 1090 | ACB, CA3, CP, LGd, OT, SI, SNr, TH, ZI, root                           |
| Moniz_2017-05-15     | M   | 210 days | Ai95 x VGlut1-Cre   | 3229         | 247    | 806   | 606  | APN, CA1, DG, LP, POL, SUB, VISam, VISpm, root                         |
| Moniz_2017-05-16     | M   | 210 days | Ai95 x VGlut1-Cre   | 3037         | 235    | 1144  | 899  | CA1, DG, LGd, LP, MG, SPF, SUB, TH, VISa, VISp, root                   |
| Moniz_2017-05-18     | M   | 210 days | Ai95 x VGlut1-Cre   | 2761         | 124    | 795   | 578  | LD, MOp, SSp, VAL, root                                                |
| Muller_2017-01-07    | F   | 119 days | Ai95 x VGlut1-Cre   | 4078         | 444    | 1109  | 646  | ACA, ACB, IC, MOs, MRN, PL, RSP, SCsg, TT, VISp, root                  |
| Muller_2017-01-08    | F   | 119 days | Ai95 x VGlut1-Cre   | 2820         | 151    | 1455  | 1270 | CA1, DG, LP, PO, VISam, root                                           |
| Muller_2017-01-09    | F   | 119 days | Ai95 x VGlut1-Cre   | 2983         | 187    | 1426  | 1364 | CA1, CA2, MRN, RSP, SCig, SCsg, VISpm, VISrl, root                     |
| Radnitz_2017-01-08   | F   | 140 days | Snap25-GCaMP6s      | 3319         | 261    | 1266  | 885  | ACA, DP, ILA, MOs, MRN, PL, RSP, SCig, SCsg, TT, VISp, root            |
| Radnitz_2017-01-09   | F   | 140 days | Snap25-GCaMP6s      | 3028         | 178    | 1161  | 1056 | MOs, MRN, SCig, VISp, root                                             |
| Radnitz_2017-01-10   | F   | 140 days | Snap25-GCaMP6s      | 3657         | 253    | 1251  | 565  | ACA, ACB, CA1, DG, LD, LP, MOs, PL, PO, VISam, root                    |
| Radnitz_2017-01-11   | F   | 140 days | Snap25-GCaMP6s      | 2847         | 142    | 860   | 549  | APN, CA1, DG, LP, MRN, POL, RSP, SCig, SCsg, VISpm, root               |
| Radnitz_2017-01-12   | F   | 140 days | Snap25-GCaMP6s      | 3156         | 128    | 1147  | 750  | CP, MOp, SSp, root                                                     |
| Richards_2017-10-29  | M   | 182 days | C57BL/6J            | 2648         | 143    | 1461  | 942  | ACA, CA1, CL, DG, LP, LS, MD, MOs, PO, RSP, SUB, VISa, VPM, root       |
| Richards_2017-10-30  | M   | 182 days | C57BL/6J            | 2999         | 237    | 1251  | 979  | CA3, MOs, OLF, ORB, POST, SCm, SNr, TH                                 |
| Richards_2017-10-31  | M   | 182 days | C57BL/6J            | 2968         | 260    | 778   | 522  | MOs, MRN, OLF, ORB, PAG, RSP, SCm, SCs, root                           |
| Richards_2017-11-01  | M   | 182 days | C57BL/6J            | 2761         | 191    | 854   | 613  | CA1, CA3, CP, EP, LGd, SNr, VPM, ZI, root                              |
| Richards_2017-11-02  | M   | 182 days | C57BL/6J            | 3242         | 296    | 928   | 781  | BMA, COA, LP, MB, PO, RT, root                                         |
| Tatum_2017-12-06     | F   | 105 days | Vglut1-Cre          | 3432         | 311    | 1425  | 795  | ACA, CA1, DG, LGd, LH, LP, LS, MD, MOs, RSP, TH, VISam, root           |
| Tatum_2017-12-07     | F   | 105 days | Vglut1-Cre          | 3203         | 258    | 1409  | 682  | CA, MOs, MRN, PL, SCm, SCsg, VISam, root                               |
| Tatum_2017-12-08     | F   | 105 days | Vglut1-Cre          | 3339         | 181    | 1907  | 1057 | MB, MOs, ORB, PAG, RSP, SCm, SCs, VISam, root                          |
| Tatum_2017-12-09     | F   | 105 days | Vglut1-Cre          | 2987         | 199    | 1041  | 805  | BLA, CA3, GPe, LGd, SNr, root                                          |
| Theiler_2017-10-11   | F   | 322 days | Ai95 x VGlut1-Cre   | 3222         | 343    | 1793  | 1348 | ACA, CA1, DG, ILA, LP, MOs, PL, SUB, VISam, VISp, root                 |

## 6. Behavioural performance per session

| Session              | Trials | Included | % rewarded | % no-go | Median response – go cue (s) |
|----------------------|--------|----------|------------|---------|------------------------------|
| Cori_2016-12-14      | 214.0  | 201.0    | 65.9       | 34.6    | 0.521                        |
| Cori_2016-12-17      | 251.0  | 229.0    | 63.3       | 37.5    | 0.509                        |
| Cori_2016-12-18      | 228.0  | 216.0    | 66.2       | 33.8    | 0.441                        |
| Forssmann_2017-11-01 | 249.0  | 241.0    | 66.7       | 50.6    | 1.797                        |
| Forssmann_2017-11-02 | 254.0  | 246.0    | 66.1       | 40.9    | 0.470                        |
| Forssmann_2017-11-04 | 290.0  | 288.0    | 74.1       | 31.7    | 0.378                        |
| Forssmann_2017-11-05 | 252.0  | 244.0    | 67.1       | 24.6    | 0.385                        |
| Hench_2017-06-15     | 250.0  | 223.0    | 64.4       | 37.2    | 0.461                        |
| Hench_2017-06-16     | 372.0  | 350.0    | 68.5       | 36.0    | 0.309                        |
| Hench_2017-06-17     | 447.0  | 430.0    | 62.0       | 34.7    | 0.355                        |
| Hench_2017-06-18     | 342.0  | 334.0    | 79.5       | 28.4    | 0.337                        |
| Lederberg_2017-12-05 | 340.0  | 327.0    | 73.8       | 18.8    | 0.179                        |
| Lederberg_2017-12-06 | 300.0  | 291.0    | 79.7       | 18.7    | 0.192                        |
| Lederberg_2017-12-07 | 268.0  | 256.0    | 69.4       | 22.8    | 0.270                        |
| Lederberg_2017-12-08 | 404.0  | 397.0    | 76.5       | 24.0    | 0.265                        |
| Lederberg_2017-12-09 | 280.0  | 265.0    | 71.8       | 26.4    | 0.339                        |
| Lederberg_2017-12-10 | 224.0  | 213.0    | 83.0       | 23.2    | 0.260                        |
| Lederberg_2017-12-11 | 316.0  | 311.0    | 78.2       | 24.1    | 0.186                        |
| Moniz_2017-05-15     | 247.0  | 225.0    | 57.5       | 45.3    | 1.139                        |
| Moniz_2017-05-16     | 235.0  | 225.0    | 59.1       | 40.0    | 0.635                        |
| Moniz_2017-05-18     | 124.0  | 109.0    | 53.2       | 45.2    | 0.730                        |
| Muller_2017-01-07    | 444.0  | 441.0    | 70.0       | 25.9    | 0.340                        |
| Muller_2017-01-08    | 151.0  | 141.0    | 67.5       | 43.0    | 0.768                        |
| Muller_2017-01-09    | 187.0  | 169.0    | 58.3       | 44.4    | 0.696                        |
| Radnitz_2017-01-08   | 261.0  | 253.0    | 72.0       | 29.1    | 0.240                        |
| Radnitz_2017-01-09   | 178.0  | 160.0    | 73.0       | 28.1    | 0.240                        |
| Radnitz_2017-01-10   | 253.0  | 242.0    | 62.8       | 32.0    | 0.415                        |
| Radnitz_2017-01-11   | 142.0  | 131.0    | 59.9       | 44.4    | 0.851                        |
| Radnitz_2017-01-12   | 128.0  | 112.0    | 58.6       | 50.0    | 1.500                        |
| Richards_2017-10-29  | 143.0  | 121.0    | 50.3       | 55.9    | 1.500                        |
| Richards_2017-10-30  | 237.0  | 219.0    | 69.2       | 32.1    | 0.339                        |
| Richards_2017-10-31  | 260.0  | 248.0    | 71.5       | 33.1    | 0.327                        |
| Richards_2017-11-01  | 191.0  | 174.0    | 67.5       | 29.8    | 0.290                        |
| Richards_2017-11-02  | 296.0  | 273.0    | 59.5       | 47.6    | 1.125                        |
| Tatum_2017-12-06     | 311.0  | 299.0    | 77.2       | 25.4    | 0.245                        |
| Tatum_2017-12-07     | 258.0  | 245.0    | 72.9       | 35.3    | 0.590                        |
| Tatum_2017-12-08     | 181.0  | 170.0    | 72.9       | 35.9    | 0.486                        |
| Tatum_2017-12-09     | 199.0  | 182.0    | 65.8       | 36.7    | 0.536                        |
| Theiler_2017-10-11   | 343.0  | 337.0    | 71.7       | 32.1    | 0.446                        |

## 7. Data files and columns

| File                                 | Rows                       | Columns                                                                                                                                                                                                                   |
|--------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tables/sessions.csv                  | 39                         | session_id, file, subject_id, sex, age, genotype, strain, session_start_time, duration_s, n_trials, n_units, n_good_units, regions, spontaneous_duration_s                                                                |
| tables/units.csv                     | 48,661                     | session_id, unit_id, region, probe, ccf_ap, ccf_dv, ccf_lr, peak_channel, depth_um, phy_annotation, good, waveform_duration_ms, waveform_duration_source, n_spikes, mean_rate_hz, task_rate_hz, spontaneous_rate_hz       |
| tables/trials.csv                    | 10,050                     | session_id, trial_id, start_time, stop_time, included, go_cue, visual_stimulus_time, visual_stimulus_left_contrast, visual_stimulus_right_contrast, response_time, response_choice, feedback_time, feedback_type, rep_num |
| tables/region_summary.csv            | 72                         | region, n_units, n_sessions, spont_rate_mean_hz, spont_rate_sd_hz, spont_rate_median_hz, mean_rate_mean_hz, waveform_duration_ms_median                                                                                   |
| trial_unit_counts.parquet            | trials × units per session | session_id, trial_id, unit_id, count_prestim, count_stim, count_remove, count_feedback                                                                                                                                    |
| spikes/<session>.parquet             | 442,033,914 total          | unit_id, time (s), amp, depth (μm)                                                                                                                                                                                        |
| behavior/<session>__<signal>.parquet | per signal                 | wheel_position, wheel_moves (start_time, stop_time, type), lick_times, eye_area, eye_xy_positions, face_motion_energy                                                                                                     |

Keys: session\_id joins every table; (session\_id, unit\_id) joins units ↔ spikes ↔ trial counts; (session\_id, trial\_id) joins trials ↔ trial counts. All times are seconds on the session clock. Column descriptions: dandi\_export/README.md.